Nonlinear Acrobot Model

1. Coordinates and Parameters

Refer Fig. 1 from the handout. The generalized coordinates are $q = [q_1,\ q_2]^T$, where q_1 is measured from the vertically downward direction and q_2 is the angle of link 2 relative to link 1. The elbow torque is τ, entering through $B = [0,\ 1]^T$. The links are uniform rods, so $l_{c1} = l_1/2$ and $l_{c2} = l_2/2$; I_1, I_2 are inertias about the link centres of mass.

$$m_1 = m_2 = 1 \text{ kg}, \quad l_1 = 1 \text{ m}, \quad l_2 = 2 \text{ m}, \quad I_1 = 0.083 \text{ kg m}^2, \quad I_2 = 0.33 \text{ kg m}^2, \quad g = 9.81 \text{ m/s}^2.$$

2. Non-linear Dynamics Equation

$$\boxed{M(q)\ddot{q} + C(q,\dot{q})\dot{q} = \tau_g(q) + B\tau}$$

$$M(q) = \begin{bmatrix} I_1 + I_2 + \dfrac{m_1 l_1^2}{4} + m_2\left(l_1^2 + \dfrac{l_2^2}{4} + l_1 l_2 \cos q_2\right) & I_2 + m_2\left(\dfrac{l_2^2}{4} + \dfrac{l_1 l_2}{2}\cos q_2\right) \\ I_2 + m_2\left(\dfrac{l_2^2}{4} + \dfrac{l_1 l_2}{2}\cos q_2\right) & I_2 + \dfrac{m_2 l_2^2}{4} \end{bmatrix}$$

$$C(q,\dot{q}) = \begin{bmatrix} -m_2 l_1 l_2 \sin q_2\, \dot{q}_2 & -\dfrac{m_2 l_1 l_2}{2}\sin q_2\, \dot{q}_2 \\ \dfrac{m_2 l_1 l_2}{2}\sin q_2\, \dot{q}_1 & 0 \end{bmatrix}$$

$$U(q) = -\left(\frac{m_1 l_1}{2} + m_2 l_1\right) g \cos q_1 - \frac{m_2 l_2}{2} g \cos(q_1 + q_2),$$

$$\tau_g(q) = -\frac{\partial U}{\partial q} = \begin{bmatrix} -\left(\dfrac{m_1 l_1}{2} + m_2 l_1\right) g \sin q_1 - \dfrac{m_2 l_2}{2} g \sin(q_1 + q_2) \\ -\dfrac{m_2 l_2}{2} g \sin(q_1 + q_2) \end{bmatrix}.$$

3. State-Space Form

With $x = [q_1,\ q_2,\ \dot{q}_1,\ \dot{q}_2]^T$,

$$\dot{x} = \begin{bmatrix} \dot{q} \\ M(q)^{-1}\big(\tau_g(q) + B\tau - C(q,\dot{q})\dot{q}\big) \end{bmatrix}, \qquad |\tau| \le 20 \text{ N m}.$$

In simulation, solve $M(q)\ddot{q} = \tau_g + B\tau - C\dot{q}$ directly rather than forming M^{-1}.

4. Project Tasks

1. Determine the equilibrium configurations and corresponding equilibrium torques.
2. Linearize the nonlinear state-space model about each equilibrium.
3. Derive the transfer functions from τ to q_1 and q_2.